

Demonstrating OpenEarable 2.0: An AI-Powered Ear Sensing Platform

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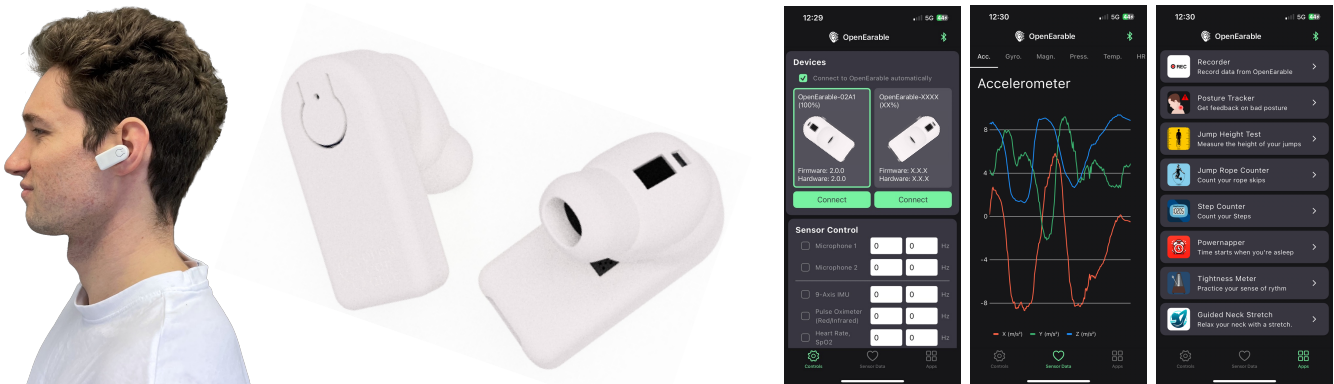


Figure 1: Overview of OpenEarable 2.0: (Left) A user wearing OpenEarable 2.0 device. (Center) Close-up images of OpenEarable 2.0 device. (Right) Screenshots of OpenEarable 2.0 mobile application, showing real-time sensor data visualization and AI applications for, e.g., fitness and health monitoring.

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Abstract

In this demo, we present OpenEarable 2.0, an open-source earphone platform designed to provide an interactive exploration of physiological ear sensing and the development of AI applications. Attendees will have the opportunity to explore real-time sensor data and understand the capabilities of OpenEarable 2.0's sensing components. OpenEarable 2.0 integrates a rich set of sensors, including two ultrasound-capable microphones (inward/outward), a 3-axis ear canal accelerometer/bone conduction microphone, a 9-axis head

inertial measurement unit, a pulse oximeter, an optical temperature sensor, an ear canal pressure sensor, a microSD slot, and a microcontroller. Participants will be able to try out the web-based dashboard and mobile app for real-time control and data visualization. Furthermore, the demo will show different applications and real-time data based on OpenEarable 2.0 across physiological sensing and health monitoring, movement and activity tracking, and human-computer interaction.

CCS Concepts

• **Hardware** → **Emerging technologies**; • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; **Ubiquitous and mobile devices**;

Keywords

earphones; in-ear; earables; hearables; open-source; open hardware; open wearables; hearing aid

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1 Introduction

Earphones have evolved from simple audio devices to sophisticated, sensor-rich platforms known as earables [4, 7]. The ear provides an ideal platform for capturing data from the skeletal, muscular, nervous, cardiovascular, respiratory, and digestive systems. Earables leverage the unique position of the ear to measure various physiological signals such as heart rate [3], respiration [9], body temperature [5], or movement [10]. While commercial earphones focus on audio applications, they lack the extensibility and capabilities necessary for broader scientific applicability.

To address this gap, we have introduced OpenEarable 2.0, an open-source earphone platform designed to empower researchers and developers by providing a comprehensive set of sensors within a wearable and ergonomic form factor. OpenEarable 2.0 integrates multiple sensing modalities, including audible-/infra-/ultrasound-capable microphones, an inertial measurement unit (IMU), a pulse oximeter, an optical temperature sensor, an ear canal pressure sensor, a microSD slot, and a microcontroller. This unique combination allows researchers to explore a wide range of physiological and behavioral phenomena.

The primary goal of this demo is to offer attendees a hands-on experience with OpenEarable 2.0, showcasing its real-time data collection capabilities, sensor fusion potential, and ease of use. The interactive demonstration highlights various AI-based applications developed on the platform. In addition to showcasing the hardware capabilities, the demo introduces the accompanying software ecosystem, including a web-based dashboard and a mobile application for data visualization and interaction. Attendees are invited to learn how to configure sensors, visualize data in real time, and even get advice on how to develop their own applications using the open-source tools provided. Through OpenEarable 2.0, we aim

to lower the entry barriers for wearable computing research and enable new opportunities for interdisciplinary collaborations in health, fitness, and interaction design.

2 Open Source Principles

OpenEarable 2.0 was designed as an open source project to promote transparency, collaboration, and innovation. The platform follows the *Guidelines of the Open Source Hardware Association (OSHW)* [6]. This entails that all hardware design files, firmware, and software are publicly and freely accessible. The platform is designed to be easily expanded and modified even by inexperienced users. It uses standard components that are readily available and inexpensive. Intuitive and accessible assembly instructions allow anyone to reproduce and improve OpenEarable 2.0.

3 Hardware

OpenEarable 2.0 hardware is designed to provide a versatile and extensible ear-based sensing platform (see Figure 2). At its core, OpenEarable 2.0 features the *Nordic Semiconductor nRF5340 (RAY-TAC MDBT53)* microcontroller, a dual-core *Arm® Cortex®-M33* processor that supports Bluetooth 5.4, including Bluetooth Low Energy (BLE) and LE Audio for sensor and audio streaming, respectively.

The sensing capabilities of OpenEarable 2.0 are enabled by a variety of state-of-the-art components. The *Bosch BMX160* 9-axis inertial measurement unit (IMU) integrates a 3-axis accelerometer, a 3-axis gyroscope, and a 3-axis magnetometer, allowing for accurate tracking of head movements and orientation. To monitor cardiovascular health, the device includes an *Analog Devices MAXM86161* pulse oximeter, capable of measuring heart rate and blood oxygen saturation (SpO₂) through photoplethysmography (PPG). Additionally, a *Melexis MLX90632* optical temperature sensor provides non-contact skin temperature measurements with medical-grade accuracy. For in-ear pressure sensing, the *Bosch BMP388* pressure sensor enables applications such as tracking ear canal deformation. OpenEarable 2.0 is also equipped with two *Knowles SPH0641LU4H-1* infra-/audible-/ultrasound-capable microphones, positioned inward and outward to capture acoustic signals both inside and outside the ear, facilitating applications such as voice analysis and occlusion effect-based heart rate monitoring. Moreover, OpenEarable 2.0 works like regular Bluetooth earphones via *Bluetooth LE Audio* for high-quality audio transmission. The device charges via USB-C. A 14-pin header connector allows for simple extension of additional sensors. When fully charged, the runtime is about 8 hours. The device weighs less than 9 grams and the dimensions of the device are approximately 35.5 × 16.5 × 7.5 mm.

4 Software

OpenEarable 2.0 runs on *Zephyr OS*, an open-source real-time operating system providing power management, sensor integration, and communication capabilities. Zephyr OS's modular nature enables rapid development and customization for specific research and application needs. The software ecosystem of OpenEarable 2.0 includes a web-based dashboard and a cross-platform mobile application that allows users to configure sensors, visualize real-time data, and develop custom applications. Researchers and developers

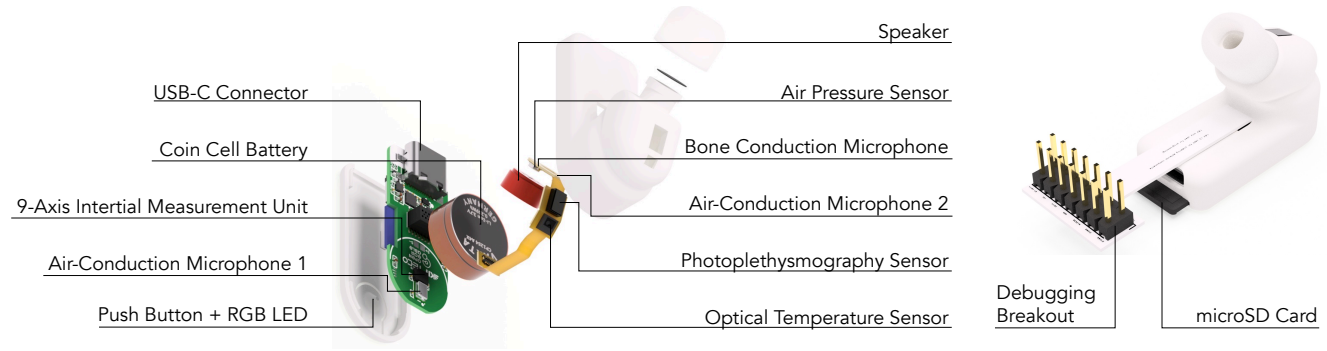


Figure 2: The front side (left) features an optical temperature sensor, photoplethysmography sensor, an air-conduction microphone, an air pressure sensor, and a bone-conduction microphone. The rear side (middle) accommodates the external air-conduction microphone, a push button with an RGB LED, a 9-axis inertial measurement unit, a USB-C connector for charging and data transfer, a microSD slot for local storage, and a 14-pin header connector for programming, debugging, and expandability.

can extend the platform through available hardware interfaces and software APIs.

5 Demonstration

The demo of OpenEarable 2.0 will provide an interactive, hands-on experience for attendees to explore its capabilities through real-time sensor data visualization and AI application testing. Overall, the demo aims to showcase the accessibility, versatility, and scientific potential of OpenEarable 2.0 in an engaging and informative way.

5.1 Real-Time Sensor Data

Attendees will explore the data streams from various sensors in real time. The dashboard will visualize key physiological parameters and motion-based metrics such as:

- **Heart Rate and SpO2:** Based on the pulse oximeter, heart rate and blood oxygen levels will be analyzed.
- **Temperature Monitoring:** The optical temperature sensor will continuously measure skin temperature.
- **Motion Tracking:** The 9-axis IMU data will illustrate head movements, orientation, and acceleration.
- **Pressure Changes:** The ear canal pressure sensor will detect subtle pressure changes related to jaw movement.
- **Acoustic Sensing:** The microphones will capture environmental and body-internal acoustic signals.
- **Audio Replay:** The earphones will play binaural (left and right) audio streamed via LEAudio.

Attendees will be encouraged to perform simple movements such as nodding, shaking their heads, or deep breathing to observe real-time responses on the dashboard. They will be able to see their real-time sensor data on a smartphone app. Via the dashboard, they can configure the sensors, e.g., adjust their sampling rates. A 3D model will show the OpenEarable device rotating in sync with the movement of the IMU to illustrate head movements. We will also demonstrate data recording and labeling for further analysis. Participants will also be encouraged to use OpenEarable 2.0 as an interactive device.

Interaction modalities that will be directly reflected by sensor data are:

- **Hand-based Input:** A well established gesture method for earable devices is device tapping (e.g., [2]). OpenEarable 2.0 can show sensor data related to tapping movements on the device using motion sensors.
- **Mouth-based Input:** Related research has demonstrated mouth-related gesture input with earphones that can be captured by the motion or pressure sensors which participants can explore during the demo [1].
- **Tensor Tympani Input:** Approximately 30% of the population can voluntarily contract the tensor tympani muscle in the middle ear [8]. Sensors inside OpenEarable 2.0 can measure the contractions of the muscle.

5.2 AI Applications

Participants will have the opportunity to interact with a variety of pre-developed applications designed to demonstrate the versatility of OpenEarable 2.0. Some of the available applications include:

- **Posture Tracker:** Monitors head position and movement patterns to ensure proper posture maintenance.
- **Guided Neck Stretching:** Guides the user through a neck stretching exercise.
- **Rope Skipping Counter:** Counts the number of successful rope skips based on head motion detection.
- **Step Counter:** Tracks step count during walking and running activities.
- **Tightness Meter:** Detects synchronization with music beats by analyzing nodding patterns.
- **Nap Timer:** Detects if the user is moving to precisely time the duration of a nap.
- **Jump Height Test:** Utilizes the IMU data to measure jump height and assess performance.

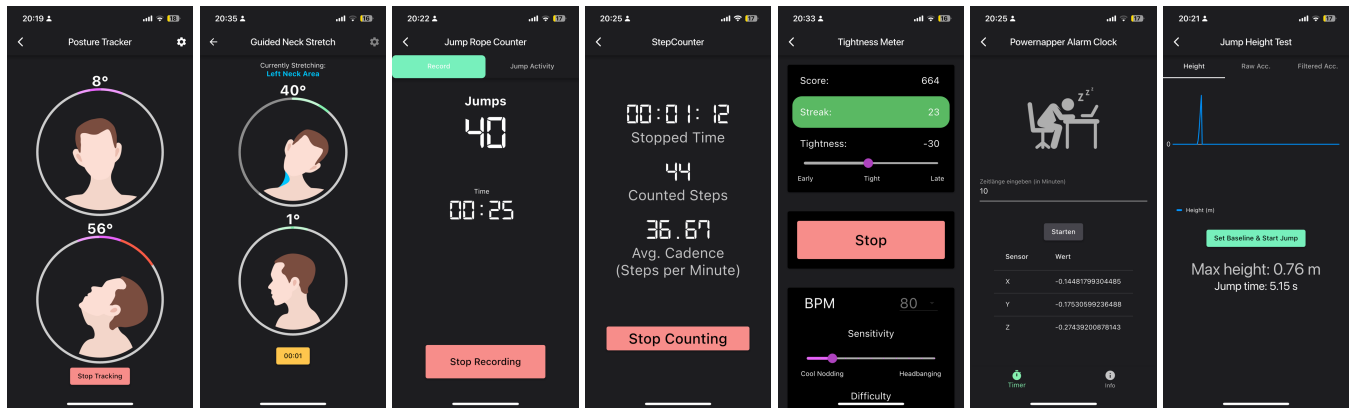


Figure 3: The applications from left to right are: (i) a posture tracker; (ii) a guided neck stretching app; (iii) a jump rope counter; (iv) a step counter; (v) a tightness meter for head nodding; (vi) a nap timer; (vii) a jump height measurement test.

6 Conclusion

OpenEarable 2.0 represents a significant advancement in the field of wearable technology by providing an open-source, sensor-rich earphone platform that supports a wide range of applications in health monitoring, activity tracking, and human-computer interaction. It integrates a variety of sensors in a miniaturized, portable format, enabling the detection of up to 30 different phenomena occurring in, on, or around the ears. The open-source nature of the hardware and software is designed to encourage collaboration within the research community and to develop new applications for wearable technology. Upon completion of the demonstration, participants will possess a profound understanding of the multifaceted sensing capabilities of OpenEarable 2.0 and the methods by which to access open-source hardware and software resources for the development of their own research projects. We hope that the platform can serve as foundation for further innovation in wearable technology and help to bridge the gap between research and real-world applications.

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